L22 20/10/2020

Isomorphism: If $\mathcal{V}$ and $\mathcal{W}$ are two vector spaces and $\tau : \mathcal{V} \longrightarrow \mathcal{W}$ such that

1. τ is one-to-one and onto
2. τ preserves the structure of the vector space.

$$\tau(v_1 + v_2) = \tau(v_1) + \tau(v_2) \qquad v_1, v_2 \in \mathcal{V}$$
$$\tau(rv) = r\,\tau(v) \qquad v \in \mathcal{V},\ r \in \mathbb{F}$$

$\mathcal{V}$ is isomorphic to $\mathcal{W}$

then τ is called an isomorphism of $\mathcal{V}$ onto $\mathcal{W}$.

e.g. $\mathcal{P}_2 = \{ p(x) \in \mathcal{P}_2 \mid p(x) = a_0 + a_1 x + a_2 x^2 \text{ where } a_0, a_1, a_2 \in \mathbb{R} \}$

$$\mathbb{R}^3 = \left\langle \begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix} \right\rangle$$

$$\tau : \mathcal{P}_2 \longrightarrow \mathbb{R}^3$$

such that $\tau(a_0 + a_1 x + a_2 x^2) = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$ $\qquad \mathcal{P}_2 \cong \mathbb{R}^3$

Two normed spaces $\mathcal{V}_1$ and $\mathcal{V}_2$ are said to be linearly isometric, denoted $\mathcal{V}_1 \cong \mathcal{V}_2$ if $\exists$ a linear operator $\tau : \mathcal{V}_1 \longrightarrow \mathcal{V}_2$ such $\|\tau v\| = \|v\| \quad \forall\, v \in \mathcal{V}_1$.

τ is called an isometric isomorphism.

$\mathcal{L}_2(j\mathbb{R})$ space: $\mathcal{L}_2(j\mathbb{R})$ or $\mathcal{L}_2$ is a Hilbert space of matrix-valued (or scalar valued) functions on $j\mathbb{R}$ consisting of all complex-valued functions F such that the integral below is bounded:

$$\int_{-\infty}^{\infty} \operatorname{trace}\left[F^*(j\omega)\, F(j\omega)\right] d\omega \underline{\underline{< \infty}} \leftarrow$$

The inner product is defined as

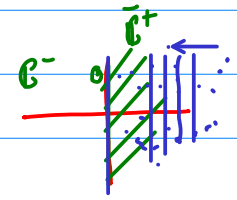
$$\langle F, G \rangle := \frac{1}{2\pi} \int_{-\infty}^{\infty} \operatorname{trace}\left[F^*(j\omega)\, G(j\omega)\right] d\omega < \infty \qquad \forall\, F, G \in \mathcal{L}_2(j\mathbb{R})$$

and the norm induced $\|F\| := \sqrt{\langle F, F \rangle}$.

e.g. $G(j\omega) = 1/(1 + j\omega) \in \mathcal{L}_2(j\mathbb{R})$ $\displaystyle\int_{-\infty}^{\infty} \left(1/(1+\omega^2)\right) d\omega = \tan^{-1}\omega \Big|_{-\infty}^{\infty} = \pi < \infty$

All <u>real, rational, strictly proper transfer</u> matrices with no poles on the imaginary axis forms a subspace of $\mathcal{L}_2(j\mathbb{R})$. This is denoted by $R\mathcal{L}_2(j\mathbb{R})$ or $R\mathcal{L}_2$.

e.g. $G(s) = 1/s+1 \in R\mathcal{L}_2$



H_2-space: A function $U: \bar{\mathbb{C}}^+ \longrightarrow \mathbb{C}^{n\times m}$ is in H_2 if

Hilbert space

(a) $U(s)$ is analytic on $\mathbb{C}^+$. ✓

(b) For almost every real number w

$$\lim_{\sigma \to 0^+} U(\sigma + jw) = U(jw) \quad ✓$$

$H_2 \subseteq \mathcal{L}_2(j\mathbb{R})$

(c) $$\sup_{\sigma \geq 0} \int_{-\infty}^{\infty} \text{trace}\left[U^*(\sigma+jw)\,U(\sigma+jw)\right] dw < \infty \iff \int_{-\infty}^{\infty} \text{trace}\left[U^*(jw)\,U(jw)\right] dw < \infty \quad ✓$$

Maximum modulus theorem: If $f(s)$ is defined and continuous on a closed-bounded set S and analytic on the interior of S, then the maximum of $|f(s)|$ on S is attained on the boundary of S.

$$\max_{s\in S} |f(s)| = \max_{s \in \partial S} |f(s)|$$

$H_2 \subseteq \mathcal{L}_2$: $F \in H_2$ $\quad \|F\| = \sqrt{\langle F, F\rangle}$

$$\sup_{\sigma>0} \frac{1}{2\pi}\int_{-\infty}^{\infty} \text{trace}\left[F^*(\sigma+jw)\,F(\sigma+jw)\right] dw$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \text{trace}\left[F^*(jw)\,F(jw)\right] dw$$

e.g. $G(s) = 1/s+1$

$G(s) = e^s/s+1 \notin H_2$ $\quad \sup_{\sigma>0} \frac{1}{2\pi}\int_{-\infty}^{\infty} \left(\frac{\boxed{e^{\sigma}}\,e^{-jw}}{\sigma - jw + 1}\right)\left(\frac{e^{\sigma+jw}}{\sigma+jw+1}\right) dw \not< \infty$

The real, rational, strictly proper transfer matrices that are <u>stable</u> forms RH_2.

$RH_2 \subseteq H_2$.

$H_2^\perp$-space: $H_2^\perp$ is the orthogonal complement of H_2. $\quad RH_2^\perp \leftarrow$ poles are in $\mathbb{C}^+$

$\mathcal{L}_2(j\mathbb{R})$, $\quad H_2 \subseteq \mathcal{L}_2(j\mathbb{R})$, $\quad H_2^\perp \subseteq \mathcal{L}_2(j\mathbb{R})$ $\qquad \mathcal{L}_2 = H_2 + H_2^\perp$

e.g. If $G(s) \in H_2$, then $G^T(-s) \in H_2^\perp$

$1/s+1 \qquad 1/-s+1 \in RH_2^\perp \subseteq H_2^\perp$

$f \in \mathcal{L}_2(-\infty, \infty)$

Bilateral Laplace transform: $\int_{-\infty}^{\infty} f(t) e^{-st}\, dt =: F(s) = \Lambda(f(t))$

$\Lambda^{-1}(F(s)) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(s) e^{st}\, dt$

$$\Lambda : \mathcal{L}_2(-\infty, \infty) \longmapsto \mathcal{L}_2(j\mathbb{R})$$

s.t. $\Lambda(f(t)) := \int_{-\infty}^{\infty} f(t) e^{-st}\, dt.$

Λ is a isometric isomorphism.

$$\langle f(t), g(t) \rangle = \langle \Lambda(f(t)), \Lambda(g(t)) \rangle$$
$$= \langle F(j\omega), G(j\omega) \rangle$$
$$= \langle F(s), G(s) \rangle$$

$$\boxed{\mathcal{L}_2(-\infty, \infty) \cong \mathcal{L}_2(j\mathbb{R})}$$

$$\boxed{\mathcal{L}_2[0, \infty) \cong \mathcal{H}_2} \qquad \boxed{\mathcal{L}_2(-\infty, 0] \cong \mathcal{H}_2^{\perp}}$$